

AI4DFT

Density Functional Theory from First Principles to AI
A beginner-to-advanced course for materials engineers

Companion to the AI4DFT project: read each chapter here, run the matching lesson in `lessons/` yourself, then complete the tasks in `CURRICULUM.md`. The interactive site (`index.html`) animates the key ideas.

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Chapter 0 — Why DFT Exists

Every property you handled in your materials engineering degree — Young's modulus, yield behavior's electronic underpinnings, band gaps, phase diagrams, diffusion barriers, corrosion potentials — is ultimately fixed by one thing: how electrons arrange themselves around atomic nuclei. Quantum mechanics (1926) wrote down the exact rule for that arrangement, the Schrödinger equation. So in principle, materials science ended in 1926: solve the equation, read off the material.

In practice, the equation is unsolvable for anything bigger than a few electrons, and Chapter 2 shows precisely why — the object it asks for grows exponentially with electron count. For four decades the field survived on cleverness and approximation around the edges.

Density Functional Theory (Hohenberg & Kohn 1964, Kohn & Sham 1965; Nobel Prize in Chemistry 1998) changed the object of the game. Instead of the many-electron wavefunction $\Psi(r_1, \dots, r_N)$ — a function of $3N$ coordinates — DFT works with the electron density $n(r)$, a function of just three coordinates no matter how many electrons the material has. Two theorems guarantee this is legal; one engineering construction (Kohn-Sham) makes it computable; one approximated term (exchange-correlation) carries all the remaining error.

Today DFT underlies roughly 40,000 papers per year, industrial R&D in semiconductors, batteries, catalysts and alloys, and the big materials databases (Materials Project, OQMD, AFLOW). Those databases in turn became training data for the machine-learning models — CHGNet, MACE, GNoME — that now deliver near-DFT accuracy about ten thousand times faster. That last step is the 'AI4DFT' of this course's title: AI did not replace DFT; DFT became the ground truth that AI learns from.

The course's one-line summary: DFT = (Kohn-Sham equations) + (self-consistent field loop) + (an approximation for exchange-correlation). Every chapter builds one of those pieces; the lessons make you operate each one yourself.

Chapter 1 — The Quantum Minimum

1.1 The wavefunction and the Schrödinger equation

A quantum particle has no definite position; it is described by a wavefunction $\psi(x)$, and $|\psi(x)|^2$ is the probability density of finding it at x . All of chemistry runs on the time-independent Schrödinger equation, which in atomic units (electron mass = 1, $\hbar = 1$, energies in *hartree*: 1 hartree = 27.211 eV, lengths in *bohr* = 0.529 angstrom) reads:

$$-(1/2) \psi''(x) + V(x) \psi(x) = E \psi(x)$$

Read it as a machine: the potential $V(x)$ is the landscape; solutions exist only for special energies E — the energy levels; the corresponding ψ are the stationary states (orbitals). Confinement causes quantization: squeezing a particle into a smaller region forces its kinetic energy up and spreads its allowed levels further apart. That is why nanostructures have size-dependent colors and gaps — quantum confinement is an engineering variable.

1.2 The big computational idea: it is an eigenvalue problem

Put ψ on a grid of points spaced dx apart, approximate the second derivative with the three-point stencil (the same one used for the heat equation in classical numerics),

$$\psi''(x_i) \sim (\psi_{i+1} - 2\psi_i + \psi_{i-1}) / dx^2$$

and the Schrödinger equation becomes a plain matrix equation $H \psi = E \psi$, where H (the Hamiltonian) is a tridiagonal matrix: kinetic stencil plus V on the diagonal. Eigenvalues are the allowed energies; eigenvectors are the orbitals. Every electronic-structure code on earth — VASP, Quantum ESPRESSO, Gaussian — is at heart diagonalizing a (much fancier) Hamiltonian. Lesson 1 has you do it for three classic potentials; the double well deserves special mention:

The double well is the origin story of bonding. Two wells side by side ('two atoms') split the ground level into an even (bonding) and odd (antibonding) pair, separated by a tunneling energy. Bring the wells closer: the splitting grows. Repeat the well 10^{23} times (a crystal): each level smears into a continuous band, and the splittings become band gaps. Lesson 1 panel C and Lesson 6 are the same physics at two scales.

1.3 Hydrogen: the exactly solvable atom

For one electron and a nucleus of charge Z , spherical symmetry reduces the 3D problem to a 1D radial equation with an effective potential: the Coulomb well $-Z/r$ plus a centrifugal barrier $l(l+1)/2r^2$, where $l = 0, 1, 2$ labels s, p, d character. The exact energies are

$$E_n = -Z^2 / (2 n^2) \text{ hartree}, \quad n = 1, 2, 3, \dots$$

independent of l . That degeneracy ($3s = 3p = 3d$) is an accident of the pure $-1/r$ potential. In a many-electron atom, inner electrons screen the nucleus; low- l orbitals penetrate the screening cloud and drop in energy while high- l orbitals do not. The degeneracy breaks, 4s fills before 3d, and the periodic table's structure — including the transition metals you alloyed with in your degree — falls out. Note also the Z^2 : core electrons of heavy elements are savagely bound, which is why Chapter 4 will freeze them into pseudopotentials.

1.4 The variational principle and basis sets

The most load-bearing theorem in the course: for any normalized trial wavefunction, the energy expectation value is greater than or equal to the true ground-state energy. Guessing can only err upward, so 'lower energy' is a rigorous compass. Expand the guess in a fixed basis $\psi = \sum c_i g_i$ (Gaussians in chemistry codes, plane waves in solids codes) and minimizing over the coefficients turns into the generalized eigenvalue problem

$$H c = E S c$$

with S the overlap matrix (basis functions need not be orthogonal). Lesson 3 makes the trade concrete on hydrogen: the best single Gaussian gives -0.424 hartree against the exact -0.5 (Gaussians lack the sharp cusp at the nucleus and decay with the wrong tail); three Gaussians recover -0.496. More basis = closer to truth, never past it, at higher cost. Every quantum chemistry paper quotes its basis set for exactly this reason, and plane-wave codes compress the same trade-off into a single knob, the kinetic-energy cutoff.

Chapter 2 — The Many-Electron Wall

2.1 The curse of dimensionality

For N electrons the wavefunction is $\Psi(r_1, \dots, r_N)$: one function of $3N$ coordinates. Store it on a laughably coarse grid of 10 points per axis and you need 10^{3N} numbers. Silicon (14 electrons): 10^{42} . Iron (26 electrons): 10^{78} — about the atom count of the observable universe, for one atom of iron on a grid too coarse to be useful. Exact quantum mechanics does not scale; nothing that stores Ψ can. Meanwhile the density $n(r)$ on the same grid costs 1,000 numbers whether the system has two electrons or two thousand. The entire strategic question of electronic structure is: how much physics can you keep while carrying only $n(r)$?

2.2 Born-Oppenheimer: separating nuclei from electrons

Nuclei are 1,800+ times heavier than electrons, so electrons re-equilibrate essentially instantly as nuclei move. We therefore clamp the nuclei, solve for the electrons in their field, and call the resulting energy-versus-geometry map the potential energy surface. 'Compute the energy of this structure' — the phrase used everywhere in materials modeling — means one point on that surface. Forces are its gradients; phonons its curvatures; reactions its mountain passes.

2.3 Mean-field theory, the SCF loop, and correlation

The first workaround (Hartree 1928): let each electron move in the average electrostatic field of the others. Each orbital then obeys a single-particle equation containing the Hartree potential $v_H(r)$, the classical repulsion of the total electron cloud. But the cloud depends on the orbitals you are solving for — chicken and egg. The resolution is the self-consistent field (SCF) loop, the algorithm at the heart of every electronic-structure code since:

```
guess n -> build V[n] -> solve for orbitals -> new n' -> mix: n =  
(1-beta) n + beta n' -> repeat until n stops changing
```

The mixing parameter β is not cosmetic. Take the new density whole ($\beta = 1$) and the loop can oscillate or diverge; mix too timidly and convergence crawls. Production codes expose this knob (`mixing_beta` in Quantum ESPRESSO) and stubborn systems — metals, magnetic materials — are tamed by tuning it. Lesson 4 has you break and fix this yourself on helium.

Helium is the perfect scorecard. Ignore electron repulsion entirely: $E = -4.0$ hartree, hopeless. Mean field (equivalent to Hartree-Fock here): -2.862 . Experiment: -2.904 . The missing 0.042 hartree (about 1.1 eV — a chemical bond's worth!) is the **correlation energy**: the energy of electrons actively dodging each other, which no average field can capture. Hartree-Fock adds exchange — the quantum-statistical repulsion between same-spin electrons required by the Pauli principle — but correlation remains. Wavefunction methods that chase it directly (CI, coupled cluster) scale like N^6 or worse: fine for molecules, fatal for solids. This is the cliffhanger DFT resolves.

Chapter 3 — Density Functional Theory Proper

3.1 The Hohenberg-Kohn theorems (1964)

Theorem 1 (the miracle): the ground-state density $n(r)$ uniquely determines the external potential, hence the Hamiltonian, hence *everything* — the full many-body wavefunction, the energy, every observable. No information is lost in passing from Ψ (3N coordinates) to n (3 coordinates).

Theorem 2: there exists a universal energy functional $E[n]$, and the true ground-state density minimizes it — a variational principle for the density alone. The catch: the theorems prove the functional exists without telling us its form. Pure Hohenberg-Kohn is a license, not a recipe.

3.2 The Kohn-Sham construction (1965)

Kohn and Sham supplied the recipe with a brilliant dodge: invent a fictitious system of *non-interacting* electrons, engineered so its density equals the real system's. Non-interacting electrons we can solve exactly — one single-particle equation each. All the unknown many-body physics is quarantined into one term of the effective potential:

$$\left[-\frac{1}{2} \text{Laplacian} + v_{\text{ext}}(r) + v_{\text{H}}[n](r) + v_{\text{xc}}[n](r) \right] \phi_i = \epsilon_i \phi_i, \quad n(r) = \sum_{\text{occupied}} |\phi_i|^2$$

v_{ext} is the nuclei's pull (known), v_{H} the classical cloud repulsion (computable from n), and v_{xc} — exchange-correlation — is 'everything else': Pauli exchange, correlation, and the kinetic-energy difference between real and fictitious electrons. Because the potential depends on the density it produces, the equations are solved with the SCF loop of Chapter 2. The total energy is assembled from the eigenvalues with double-counting corrections subtracted (the eigenvalue sum counts electron-electron repulsion twice; Lesson 5's code shows each correction line by line).

Lesson 5 is the course's centerpiece: a complete, working Kohn-Sham DFT code in about a hundred lines of numpy. Fed two soft nuclei and two electrons, it iterates to self-consistency, scans the bond length, and finds an equilibrium bond — a chemical bond emerging from density alone. After you have modified that code (the curriculum tasks make you delete exchange, overfill orbitals, and destabilize the mixing), production DFT is 'the same machine, bigger'.

3.3 Exchange-correlation: Jacob's ladder

All of DFT's error lives in the approximation chosen for v_{xc} . The community ranks approximations on 'Jacob's ladder' — each rung feeds the functional more information about the density, costs more, and (usually) delivers more accuracy:

Rung	Name / exemplar	Uses	Character
1	LDA (1965)	n at each point	Treats every point as uniform electron gas. Shockingly decent for solids; overbinds (~1-2% short lattice constants).

Run	Name / exemplar	Uses	Character
2	GGA — PBE (1996)	n and its gradient	THE workhorse of materials science and the Materials Project's historical default. Good geometries and energetics.
3	meta-GGA — SCAN / r2SCAN	+ kinetic-energy density	Better formation energies and diverse bonding; the new database standard.
4	Hybrids — HSE06, B3LYP	+ a fraction of exact exchange	Respectable band gaps (Si ~1.2 eV vs expt 1.17); 10-100x cost in plane-wave codes.
5	RPA, GW, ML functionals (DM21)	unoccupied states / learned forms	Frontier accuracy; GW is the standard for true quasiparticle gaps.

Memorize the two systematic failures. **(1) Band gaps:** LDA and PBE underestimate them by 40-50% (silicon: ~0.6 eV predicted vs 1.17 eV measured) — a structural flaw (the missing 'derivative discontinuity'), not sloppiness, and the reason band-structure papers reach for hybrids or GW. **(2) Van der Waals:** semilocal functionals miss London dispersion entirely; layered materials (graphite!) and molecular crystals need explicit corrections (DFT-D3, vdW functionals). Knowing which functional fails where is exactly the craft skill that separates a user from an operator.

Chapter 4 — Practical DFT on Real Materials

4.1 The three practicalities

Plane waves and the cutoff. Crystals are periodic, so Bloch's theorem labels electron states by a wavevector k in the Brillouin zone and expands orbitals in plane waves $\exp(i(k+G)r)$ — the crystal-adapted basis used by VASP, Quantum ESPRESSO, CASTEP and Abinit. The basis size is set by one knob, the kinetic-energy cutoff (`ecutwfc`): include all plane waves with $(1/2)|k+G|^2$ below it. By the variational principle, raising the cutoff only improves the energy — so you raise it until answers stop changing. That is a convergence test, and Lesson 6's task (breaking the bands by deleting plane waves) is its toy version.

k-point sampling. Crystal properties are integrals over the Brillouin zone, computed on a mesh of k-points (Monkhorst-Pack grids). Metals need dense meshes (their Fermi surface is sharp); insulators forgive coarse ones. Second convergence test: densify the mesh until answers stop changing.

Pseudopotentials. Core electrons are chemically inert but violently bound (that Z-squared from Chapter 1), and their sharp wiggles would demand absurd cutoffs. So each element's core is replaced by a smooth effective potential ('pseudopotential' / PAW dataset) and only valence electrons are solved explicitly. You will download curated pseudopotential libraries rather than make your own — but you must always report which ones you used.

4.2 The workflow layer: ASE

The community's Python glue is ASE (Atomic Simulation Environment). Its central design idea, the Calculator interface, is worth internalizing: an Atoms object holds the structure; a calculator — classical EMT, real DFT (GPAW, Quantum ESPRESSO), or a neural network (CHGNet) — is attached with one line; `get_potential_energy()` and `get_forces()` look identical regardless. You prototype the whole study with a cheap calculator, then swap the engine. Lesson 7 runs the classic first calculation this way: strain a crystal, record $E(V)$, fit a Birch-Murnaghan equation of state, and read off the equilibrium lattice constant and bulk modulus — numbers you can check against X-ray diffraction and ultrasonic measurements. On this machine real DFT engines need WSL (they are Linux-native); the curriculum's Section 4.3 walks the setup.

The iron rule: a DFT number without convergence tests (cutoff, k-mesh) and a validation anchor (experiment or higher theory) is not a result. Reviewers, and your future self, will ask for both.

4.3 What DFT computes for a materials engineer

Property	How	Used for
Lattice constants, bulk/elastic moduli	$E(V)$ and strain fits	alloy screening, stiffness, pressure response
Band structure & DOS	eigenvalues along k-paths	metal/semiconductor character, gaps, effective masses

Property	How	Used for
Formation energies & convex hulls	total-energy differences	phase stability, alloy design, battery voltages
Defect energetics	supercells with vacancies/dopants	diffusion, doping limits, radiation damage
Surfaces & adsorption	slab models	catalysis, corrosion, growth
Phonons & thermodynamics	force-constant curvatures	thermal properties, stability at temperature

Chapter 5 — AI4DFT

5.1 DFT as a data factory

Because DFT is accurate enough to trust and cheap enough to run at scale, the 2010s produced something new: databases of millions of computed materials. The Materials Project, OQMD and AFLOW each hold DFT-computed structures, energies, band gaps and elastic data for hundreds of thousands to millions of compounds, all queryable by API. This is the substrate that made machine learning possible: labeled training data at scale, with consistent physics.

5.2 Machine-learned interatomic potentials (MLIPs)

An MLIP is a neural network trained to map {atomic structure} directly to {energy, forces}, skipping the quantum mechanics at inference time. Modern ones are graph neural networks: atoms are nodes, neighbor relations are edges, and successive message-passing layers build up each atom's chemical environment before summing per-atom energies. Forces come free by automatic differentiation — the same machinery as your PDE track's autodiff chapter. 'Universal' potentials — CHGNet (trained on ~1.5 million Materials Project results, 89 elements), MACE-MP, M3GNet — cover most of the periodic table out of the box. The payoff is brutal: milliseconds per evaluation versus minutes-to-hours for DFT, a factor of about 10^4 , which converts week-long dynamics studies into lunch breaks and enables screening campaigns of millions of candidates (Google DeepMind's GNoME proposed 2.2 million new crystals, ~380k predicted stable, by iterating generation against DFT verification).

The catch, stated bluntly: **an MLIP is an interpolation of its training set**. Inside the distribution — near-equilibrium structures of database-like chemistries — it is superb. Outside — exotic stoichiometries, extreme pressures, bond breaking, surfaces it never saw — it can be confidently, silently wrong. Professional practice is spot-check validation: run real DFT on a sample of the MLIP's outputs, especially anything surprising. Lesson 8 has you relax a deliberately damaged crystal with CHGNet, time the speedup, and then — in the tasks — push it until it misbehaves.

5.3 The other AI frontiers

ML functionals: DeepMind's DM21 (2021) replaced the hand-built exchange-correlation functional with a trained neural network, beating standard functionals on hard molecular benchmarks — AI inside DFT's engine room rather than around it. **Delta learning:** train a small model on the *difference* between cheap and expensive theory, getting expensive accuracy at cheap cost. **Generative discovery:** models like GNoME and MatterGen propose novel crystal structures which DFT then verifies — the design loop your capstone imitates in miniature. In every case the pattern is the same: DFT remains the arbiter; AI multiplies its reach.

Chapter 6 — Study Guide

Read a chapter, run its lessons, do the tasks in CURRICULUM.md (checkpoint tasks are starred there), tick the progress log in README.md. The mapping:

Chapter here	You run	Task block
1 Quantum minimum	lesson01, lesson02, lesson03	CURRICULUM Section 1
2 Many-electron wall	lesson04	Section 2
3 DFT proper	lesson05 (centerpiece)	Section 3
4 Practical DFT	lesson06, lesson07 (pip install ase), then WSL+GPAW	Section 4
5 AI4DFT	lesson08 (pip install chgnet)	Section 5
Capstone	your pick: alloy hull / battery voltage / your question	Section 6

Further reading, in the order a beginner should meet them:

- Sholl & Steckel, *Density Functional Theory: A Practical Introduction* — the gentlest real book; read alongside Chapters 3-4.
- Martin, *Electronic Structure: Basic Theory and Practical Methods* — the field's reference tome; consult, don't read linearly.
- Hohenberg & Kohn, Phys. Rev. 136, B864 (1964); Kohn & Sham, Phys. Rev. 140, A1133 (1965) — the founding papers, surprisingly readable after Chapter 3.
- Perdew, Burke & Ernzerhof, Phys. Rev. Lett. 77, 3865 (1996) — PBE, the most cited physics paper of its era.
- Deng et al., *CHGNet*, Nature Machine Intelligence 5, 1031 (2023); Merchant et al., *GNoME*, Nature 624, 80 (2023) — the AI4DFT state of the art you touch in Lesson 8.

The rule that never expires — every number you ever produce gets three questions: Is it converged? Is it validated? Is it in-distribution (for anything ML)? This course made you watch all three fail on purpose. That was the point.